

THE SMALLEST DENOMINATOR NOT CONTAINED IN A UNIT FRACTION DECOMPOSITION OF 1 WITH FIXED LENGTH

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ABSTRACT. Let $v(k)$ be the smallest integer larger than 1 that does not occur among the denominators in any identity of the form

$$1 = \frac{1}{n_1} + \cdots + \frac{1}{n_k},$$

where $1 \leq n_1 < \cdots < n_k$ are pairwise distinct integers. In their 1980 monograph, Erdős and Graham asked for quantitative estimates on the growth of $v(k)$ and suggested the lower bound $v(k) \gg k!$. In this paper we give the first known improvement and show that there exists an absolute constant $c > 0$ such that the inequality

$$v(k) \geq e^{ck^2}$$

holds for all positive integers k .

1. INTRODUCTION

For a positive integer k , let S_k denote the set of all k -tuples $(n_1, \dots, n_k)$ of integers satisfying

$$1 = \frac{1}{n_1} + \frac{1}{n_2} + \cdots + \frac{1}{n_k}, \tag{1.1}$$

with $1 \leq n_1 < n_2 < \cdots < n_k$. We call an equation of the form (1.1) a *k-term unit fraction decomposition of 1*. Such sums are also known in the literature as *Egyptian fraction expansions* or simply *Egyptian fractions*. In some works, repetitions among the denominators are allowed. Throughout this paper, however, we only consider decompositions with pairwise distinct denominators. We further define

$$F(k) := |S_k|$$

and

$$D_k := \{m : m = n_i \text{ for some } (n_1, \dots, n_k) \in S_k \text{ and some } i \text{ with } 1 \leq i \leq k\}.$$

Thus, D_k is the set of all denominators that occur in at least one k -term unit fraction decomposition of 1.

In their 1980 monograph [9, p. 35], Erdős and Graham defined $v(k)$ to be the smallest integer > 1 not contained in D_k , and asked how fast $v(k)$ grows as a function of k . They remarked that the lower bound $v(k) \gg k!$ should follow from earlier work of Bleicher and Erdős [2, 3, 4], and further speculated that $v(k)$ might even grow doubly exponentially. Graham reiterated the same question in [10, p. 297], and it is now listed as Problem #293 on Bloom's Erdős Problems website [5].

For an upper bound, note that we certainly have

$$v(k) \leq |D_k| + 2 \leq kF(k) + 2. \tag{1.2}$$

2020 *Mathematics Subject Classification*. Primary 11D68; Secondary 11B75.

Key words and phrases. Egyptian fractions, growth rates.

Let $c_0 = 1.264085\dots$ denote the Vardi constant.¹ By applying the upper bound on $F(k)$ due to Elsholtz and Planitzer [7, Corollary 3], we then obtain

$$v(k) \leq c_0^{\left(\frac{1}{5} + o(1)\right)2^k},$$

as the prefactor k and the additive constant $+2$ on the right-hand side of (1.2) can both be absorbed into the $o(1)$ -term.

On the other hand, to the best of our knowledge, no lower bound on $v(k)$ exists in the literature. In fact, even extracting the lower bound $v(k) \gg k!$ from the results in [2], [3] and [4] (which Erdős and Graham in [9] and [10] claim is ‘easy to see’), does not seem straightforward to us. Regardless, the purpose of this note is to prove the following stronger bound.

Theorem 1.1. *There exists an absolute constant $c > 0$ such that*

$$v(k) \geq e^{ck^2}$$

holds for all positive integers k .

2. MAIN LOWER BOUND

As observed by Konyagin [11], we have $F(k) \leq F(k+1)$ for all $k \geq 2$, due to the identity

$$\frac{1}{n} = \frac{1}{n+1} + \frac{1}{n(n+1)}. \quad (2.1)$$

Indeed, (2.1) provides an injection from S_k to S_{k+1} , sending

$$(n_1, \dots, n_k) \in S_k$$

to

$$(n_1, \dots, n_{k-1}, n_k + 1, n_k(n_k + 1)) \in S_{k+1}.$$

By a slightly more involved argument we can even show the inclusion $D_k \subseteq D_{k+1}$.

Lemma 2.1. *For all $k \geq 2$ we have $D_k \subseteq D_{k+1}$.*

Proof. Let m be an element of D_k and let $(n_1, \dots, n_k) \in S_k$ contain m . We then aim to construct a $(k+1)$ -tuple in S_{k+1} that also contains m . As one can verify

$$D_2 = \emptyset \subseteq D_3 = \{2, 3, 6\} \subseteq \{2, 3, 7, 42\} \cup \{2, 4, 6, 12\} \subseteq D_4,$$

we may assume $k \geq 4$.

In analogy with the proof of $F(k) \leq F(k+1)$, if $m \neq n_k$, then we first of all remark

$$m \in (n_1, \dots, n_{k-1}, n_k + 1, n_k(n_k + 1)) \in S_{k+1}$$

by (2.1). Secondly, if $m = n_k \notin \{n_{k-1} + 1, n_{k-1}(n_{k-1} + 1)\}$, then we have²

$$m \in (n_1, \dots, n_{k-2}, n_k, n_{k-1} + 1, n_{k-1}(n_{k-1} + 1)) \in S_{k+1},$$

once again by (2.1). We may therefore further assume $m = n_k \in \{n_{k-1} + 1, n_{k-1}(n_{k-1} + 1)\}$ from now on.

If n_{k-1} is composite, write $n_{k-1} = ab$ with a and b integers larger than 1. Generalizing (2.1), we have the identity

$$\frac{1}{n_{k-1}} = \frac{1}{n_{k-1} + a} + \frac{1}{b(n_{k-1} + a)},$$

while

$$n_{k-1} + 1 < n_{k-1} + a < b(n_{k-1} + a) < n_{k-1}(n_{k-1} + 1).$$

¹See OEIS sequence [A076393](#) for its decimal expansion [12].

²Here, as in the rest of this proof, we slightly abuse notation by writing that a $(k+1)$ -tuple is contained in S_{k+1} , where it is actually the sorted tuple that is contained in S_{k+1} .

We therefore deduce

$$m \in (n_1, \dots, n_{k-2}, n_k, n_{k-1} + a, b(n_{k-1} + a)) \in S_{k+1}.$$

On the other hand, if $n_{k-1} = p$ is prime, we claim that $m = n_k \in \{n_{k-1} + 1, n_{k-1}(n_{k-1} + 1)\}$ implies $n_k = n_{k-1}(n_{k-1} + 1)$. To see this, assume by contradiction that $m = n_k$ is equal to $n_{k-1} + 1$ instead. We then write

$$\sum_{i=1}^k \frac{1}{n_i} = \frac{1}{p} + \sum_{\substack{1 \leq i \leq k \\ i \neq k-1}} \frac{1}{n_i} = \frac{1}{p} + \frac{A}{B} = \frac{Ap + B}{Bp},$$

where $\frac{A}{B}$ is the reduced form of the remaining sum. Now, by the assumption $n_k = n_{k-1} + 1$, we see that none of the denominators n_i with $i \neq k-1$ are divisible by p , hence $\gcd(B, p) = 1$. Thus, for the numerator we have $Ap + B \equiv B \not\equiv 0 \pmod{p}$, implying that the fraction $(Ap + B)/(Bp)$ cannot simplify to a fraction with denominator not divisible by p . In particular, this fraction is not equal to the integer 1, contradicting the assumption $(n_1, \dots, n_k) \in S_k$.

With the assumption that n_{k-1} is prime and the equality $n_k = n_{k-1}(n_{k-1} + 1)$ in mind, there are now two final cases to consider.

If $n_{k-1} \neq n_{k-2} + 1$, then we claim that

$$m \in (n_1, \dots, n_{k-3}, n_{k-1}, n_k, n_{k-2} + 1, n_{k-2}(n_{k-2} + 1)) \in S_{k+1}.$$

The sum of reciprocals of this latter tuple is equal to 1 by (2.1), so it suffices to prove that all integers in the tuple are distinct. In other words, we need to show that

$$\{n_{k-1}, n_k\} \cap \{n_{k-2} + 1, n_{k-2}(n_{k-2} + 1)\} = \emptyset.$$

One sees this by the assumption $n_{k-1} \neq n_{k-2} + 1$, the assumption that n_{k-1} is prime (hence not equal to $n_{k-2}(n_{k-2} + 1)$), and the fact that $n_k = n_{k-1}(n_{k-1} + 1) > n_{k-2}(n_{k-2} + 1)$.

If $n_{k-1} = n_{k-2} + 1$, then n_{k-2} is even and, by the assumption $k \geq 4$, larger than $n_1 \geq 2$. Hence, n_{k-2} is composite and we can write $n_{k-2} = ab$ with a and b integers larger than 1. We then see

$$m \in (n_1, \dots, n_{k-3}, n_{k-1}, n_k, n_{k-2} + a, b(n_{k-2} + a)) \in S_{k+1},$$

where

$$\{n_{k-1}, n_k\} \cap \{n_{k-2} + a, b(n_{k-2} + a)\} = \emptyset$$

follows similarly as before, in this case using the assumption $n_{k-1} = n_{k-2} + 1 \neq n_{k-2} + a$. $\square$

We now recall the following result by Vose [14]:

Lemma 2.2 ([14]). *There exists a positive integer α , a sequence³ of primes $5 \leq p_1 < p_2 < \dots$ and a sequence of positive integers $N_1 < N_2 < \dots$ defined by*

$$N_K := 4^{\alpha K^2} (p_1 p_2 \dots p_K)^2$$

for which the following holds: for all fractions $\frac{a}{b} \in (0, 1)$ there is an integer K and divisors

$$d_1, \dots, d_r, d'_1, \dots, d'_s$$

of N_K such that with $u_i := \frac{N_K}{d_i}$ and $v_j := \frac{bN_K}{d'_j}$ we have both

$$1 < u_1 < \dots < u_r < v_1 < \dots < v_s$$

and

$$r + s \leq C\sqrt{\log b}$$

³Although it is not explicitly mentioned in [14] that we may assume all primes are larger than or equal to 5, this does follow from the proof of [13, Lemma 3]. We also use ever so slightly different definitions of N_K , u_i and v_j , but these differences are mainly cosmetic.

for some absolute constant C , while

$$\frac{a}{b} = \frac{1}{u_1} + \dots + \frac{1}{u_r} + \frac{1}{v_1} + \dots + \frac{1}{v_s}.$$

The proof of Theorem 1.1 will now be based on applying Lemma 2.2 to the fraction $\frac{a}{b} = \frac{m-1}{m}$.

Proof of Theorem 1.1. By iterating Lemma 2.1 it is sufficient to show that there exists an absolute constant $c > 0$ such that for all integers $k \geq 1$ and all integers m with $1 < m < e^{ck^2}$ we have $m \in D_{k'}$ for some $k' \leq k$. That is, we need to prove that there exists a unit fraction decomposition of 1 with at most k terms that contains m as one of the denominators.

Choose $c = \min\left(\frac{\log 2}{40^2}, \frac{1}{(41C)^2}\right)$ with C as in Lemma 2.2, and let m be an integer with $1 < m < e^{ck^2}$. Without loss of generality we may then assume $k \geq 41$, as for $k \leq 40$ there are no integers m with $1 < m < e^{ck^2}$ by the definition of c . Moreover, for each integer $2 \leq m \leq 432$ we have verified (see [1] for the explicit decompositions) that 1 admits a unit fraction expansion with at most 41 terms in which m occurs as a denominator. Under the standing assumption $k \geq 41$, we may therefore assume $m > 432$ throughout the remainder of the proof. We now apply Lemma 2.2 to the fraction $\frac{a}{b} = \frac{m-1}{m}$, and there are two different cases to consider.

If m does not occur among the u_i and v_j , then

$$\frac{1}{u_1} + \dots + \frac{1}{u_r} + \frac{1}{v_1} + \dots + \frac{1}{v_s} + \frac{1}{m}$$

is a unit fraction decomposition of 1 containing m as a denominator. Moreover, the number of terms is

$$\begin{aligned} r + s + 1 &\leq C\sqrt{\log m} + 1 \\ &< C\sqrt{ck} + 1 \\ &\leq \frac{k}{41} + 1 \\ &< k. \end{aligned}$$

On the other hand, if $\frac{1}{m}$ does occur as one of the u_i or v_j , then let $t \in \{1, 2, 3\}$ be such that $m + t$ is divisible by 3 and define

$$D := \{3, 6, 9, 12, 15, 18, 24, 27, 30, 45, 54, 60, 72\}.$$

We note that all $d \in D$ are divisible by 3 and one can verify that $\sum_{d \in D} \frac{1}{d} = 1$. We now consider

$$\frac{1}{u_1} + \dots + \frac{1}{u_r} + \frac{1}{v_1} + \dots + \frac{1}{v_s} + \frac{1}{m+t} + \sum_{i=0}^{t-1} \sum_{d \in D} \frac{1}{d(m+i)(m+i+1)}. \quad (2.2)$$

Since $\sum_{d \in D} \frac{1}{d} = 1$, we first note

$$\begin{aligned} \frac{1}{m+t} + \sum_{i=0}^{t-1} \sum_{d \in D} \frac{1}{d(m+i)(m+i+1)} &= \frac{1}{m+t} + \sum_{i=0}^{t-1} \frac{1}{(m+i)(m+i+1)} \\ &= \frac{1}{m+t} + \sum_{i=0}^{t-1} \left(\frac{1}{m+i} - \frac{1}{m+i+1} \right) \\ &= \frac{1}{m}, \end{aligned}$$

as the sum telescopes. We therefore obtain that the total sum in (2.2) is equal to 1. As before, the number of terms in (2.2) satisfies

$$\begin{aligned} r + s + 1 + t|D| &\leq r + s + 40 \\ &< C\sqrt{ck} + 40 \\ &\leq \frac{k}{41} + 40 \\ &\leq k, \end{aligned}$$

by our choice of c . Therefore, to complete the proof it suffices to show that all denominators in (2.2) are distinct.

To see this, observe that every denominator in (2.2) other than the u_i and v_j is a multiple of 3. On the other hand, recall that $N_K = 4^{\alpha K^2} (p_1 p_2 \cdots p_K)^2$ with $p_i \geq 5$ for all i , so that $3 \nmid N_K$. Hence, none of these new denominators can divide N_K , and in particular none of them can coincide with any u_i . Therefore, it remains to check that the denominators $m + t$ and $d(m + i)(m + i + 1)$ are all distinct from the v_j (or equivalently, that none of them are equal to m times a divisor of N_K), and also distinct from one another.

Now, $m + t \in \{m + 1, m + 2, m + 3\}$ is not a multiple of m if $m \geq 4$, while $m + t$ is smaller than all the other terms $d(m + i)(m + i + 1)$, so the denominator $m + t$ in (2.2) is certainly unique.

If $i = 0$, then $d(m + i)(m + i + 1) = dm(m + 1)$ cannot be equal to m times a divisor of N_K , as $d(m + 1)$ is divisible by 3, whereas $3 \nmid N_K$.

To see why $d(m + i)(m + i + 1)$ is, for $1 \leq i \leq t - 1$, not a multiple of m and therefore not equal to any v_j , note that

$$\begin{aligned} \gcd(m, d(m + i)(m + i + 1)) &\leq \gcd(m, d) \gcd(m, m + i) \gcd(m, m + i + 1) \\ &= \gcd(m, d) \gcd(m, i) \gcd(m, i + 1) \\ &\leq d(t - 1)t \\ &\leq 432 \\ &< m. \end{aligned}$$

We conclude that, indeed, none of the $d(m + i)(m + i + 1)$ are equal to a v_j .

Finally, we need to check that

$$d(m + i)(m + i + 1) \neq d'(m + j)(m + j + 1)$$

for all $d, d' \in D$ and all integers $0 \leq i, j \leq t - 1$, unless $d = d'$ and $i = j$. Suppose, for a contradiction, that

$$d(m + i)(m + i + 1) = d'(m + j)(m + j + 1) \tag{2.3}$$

does hold for some $d, d' \in D$ and $0 \leq i, j \leq t - 1$. By symmetry we may assume $d \geq d'$. In fact, if $d = d'$, then equation (2.3) simplifies to

$$(m + i)(m + i + 1) = (m + j)(m + j + 1).$$

As the function $x \mapsto (m + x)(m + x + 1)$ is strictly increasing for $x \geq 0$, this would imply $i = j$. Hence, we will further assume the strict inequality $d > d'$.

Inspecting the set D , one can verify that for any distinct $d > d'$ in D we actually have

$$\frac{d}{d'} \geq \frac{10}{9}. \tag{2.4}$$

Since $t \in \{1, 2, 3\}$, we have $i, j \in \{0, 1, 2\}$, hence

$$d(m + i)(m + i + 1) \geq dm(m + 1) \tag{2.5}$$

and

$$d'(m+j)(m+j+1) \leq d'(m+2)(m+3). \quad (2.6)$$

By combining (2.3), (2.4), (2.5), and (2.6), we then obtain

$$\frac{10}{9}m(m+1) \leq (m+2)(m+3),$$

contradicting $m > 432$. This completes the proof. $\square$

3. CONCLUDING REMARKS

For integers $1 \leq a < b$, let $N(a, b)$ denote the minimal t such that there exist distinct integers $1 < n_1 < \dots < n_t$ with

$$\frac{a}{b} = \frac{1}{n_1} + \dots + \frac{1}{n_t},$$

and set $N(b) := \max_{1 \leq a < b} N(a, b)$. The problem of estimating $N(b)$ goes back to Erdős [8], where he showed

$$N(b) \ll \frac{\log b}{\log \log b}.$$

Erdős and Graham [9, p. 37] asked to improve upon this, and the determination of $N(b)$ is now recorded as Erdős Problem #304 on Bloom's Erdős Problems website [6]. The current best known upper bound is by Vose [14] and gives

$$N(b) \ll \sqrt{\log b}. \quad (3.1)$$

The stronger conjecture that we actually have

$$N(b) \ll \log \log b \quad (3.2)$$

was first suggested in [8].

It is Vose's proof of (3.1) that we use in order to prove Theorem 1.1. If, however, (3.2) holds, it seems likely that our lower bound on $v(k)$ can be improved to $e^{e^{ck}}$ instead, which would match the doubly exponential growth rate suggested by Erdős and Graham (up to the constant c in the exponent).

Conversely, any lower bound for $v(k)$ immediately yields an upper bound for $N(b-1, b)$. Indeed, $b < v(k)$ implies $b \in D_k$, so there is a k -term decomposition of 1 involving $\frac{1}{b}$. Removing this term gives a $(k-1)$ -term decomposition of $\frac{b-1}{b}$, hence $N(b-1, b) \leq k-1$. This explains why Erdős Problems #293 [5] and #304 [6] are closely connected.

ACKNOWLEDGEMENTS

The authors are grateful to Thomas Bloom for founding and maintaining the Erdős Problems website.

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